

CBCS SCHEME

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21CS733

Seventh Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025

Cryptography and Network Security

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Explain the simplified model of symmetric encryption. (06 Marks)
- b. Write a note on types of attack on encrypted message. (04 Marks)
- c. Explain Feistel encryption and decryption algorithm with neat diagram. (10 Marks)

OR

- 2 a. Using Hill cipher, perform encryption and decryption for plain text = "SAFE MESSAGES" using key = "CIPHERING" (10 Marks)
- b. Explain with neat diagram the working of DES encryption algorithm. (10 Marks)

Module-2

- 3 a. Explain RSA algorithm. Using RSA algorithm perform encryption and decryption for $p = 5$, $q = 11$, $e = 3$, $M = 9$. (10 Marks)
- b. Explain the public key crypto system with neat diagram. (10 Marks)

OR

- 4 a. With a neat diagram, explain the man-in-middle attack. (10 Marks)
- b. Explain Elgamal cryptosystem. Perform encryption and decryption using : $q = 19$, $\alpha = 10$, $K = 6$, $M = 17$, $X_A = 5$, $Y_A = 3$ (10 Marks)

Module-3

- 5 a. With relevant diagram, explain the key distribution scenario. (08 Marks)
- b. Explain with a neat diagram the control vector encryption and decryption. (06 Marks)
- c. Write a note on exchange of public key certificate. (06 Marks)

OR

- 6 a. Explain the various techniques used for distribution of public key. (10 Marks)
- b. Write notes on: (10 Marks)
 - (i) Decentralized key distribution
 - (ii) Public key distribution of secret key

Module-4

- 7 a. Explain X.509 certificate format. (10 Marks)
- b. Illustrate the block diagram of PKIX architectural model. (10 Marks)

OR

- 8 a. Describe the certification path for forward and reverse certificates of X.509. (10 Marks)
- b. Explain with a suitable diagram the overview of Kerberos. (10 Marks)

Module-5

- 9 a. Explain confidentiality and authentication for S/MIME functional flow. (10 Marks)
- b. With a neat diagram, explain ESP packet format. (10 Marks)

OR

- 10 a. Write a note on IKE formats. (10 Marks)
- b. Explain the architecture of IP/Sec architecture. (10 Marks)

Important Note : 1. On completing your answers, compulsorily draw diagonal cross lines on the remaining blank pages.
2. Any revealing of identification, appeal to evaluator and/or equations written eg, $42+8=50$, will be treated as malpractice.